

7. Cloud Pub/Sub: Experiment how Google Cloud Pub/Sub facilitate real-time messaging and communication between distributed applications.

Objectives

Demonstrate how Google Cloud Pub/Sub facilitates real-time messaging and communication between distributed applications by:

- 1. Creating a Pub/Sub topic**
 - 2. Creating a subscription**
 - 3. Publishing messages to the topic**
 - 4. Receiving messages from the subscription**
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Prerequisites

- A Google Cloud Platform (GCP) project
 - Cloud SDK (gcloud CLI) installed and configured
 - Billing enabled
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Step 1: Enable Pub/Sub API

Before using Pub/Sub, enable its API:

gcloud services enable pubsub.googleapis.com

Step 2: Create a Pub/Sub Topic

A topic acts as a communication channel where publishers send messages.

```
gcloud pubsub topics create my-topic
```

Step 3: Create a Subscription

A subscription is used by subscribers to receive messages.

```
gcloud pubsub subscriptions create  
my-subscription --topic=my-topic
```

- Messages published to **my-topic** will be delivered to **my-subscription**.
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Step 4: Publish a Message

Send a test message to the topic:

```
gcloud pubsub topics publish my-topic  
--message="Hello, Pub/Sub!"
```

Step 5: Receive Messages

Fetch messages from the subscription:

```
gcloud pubsub subscriptions pull my-subscription  
--auto-ack
```

- **--auto-ack**: Acknowledges message receipt, so it is removed from the queue.
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Step 6: Simulate Real-Time Messaging

Run the following commands in two separate terminals:

Terminal 1: Continuous Subscriber (Listening)

```
gcloud pubsub subscriptions pull my-subscription  
--auto-ack --limit=10
```

Terminal 2: Publisher (Sending Messages)

```
gcloud pubsub topics publish my-topic --message="Message 1"
```

```
gcloud pubsub topics publish my-topic --message="Message 2"
```

Observe how messages appear in the subscriber terminal in real-time.

Step 7: Clean Up Resources

To avoid billing charges, delete the topic and subscription:

```
gcloud pubsub subscriptions delete my-subscription
```

```
gcloud pubsub topics delete my-topic
```

Conclusion

- ✓ Successfully demonstrated real-time messaging with Pub/Sub.
- ✓ Messages were published by a producer and received by a consumer.
- ✓ Multiple subscribers can listen to the same topic for distributed communication.

```

raghav_cs@cloudshell:~ (ringed-enigma-427005-c8)$ gcloud services enable pubsub.googleapis.com
raghav_cs@cloudshell:~ (ringed-enigma-427005-c8)$ gcloud pubsub topics create my-topic
Created topic [projects/ringed-enigma-427005-c8/topics/my-topic].
raghav_cs@cloudshell:~ (ringed-enigma-427005-c8)$ gcloud pubsub subscriptions create my-subscription --topic=my
-topic
Created subscription [projects/ringed-enigma-427005-c8/subscriptions/my-subscription].

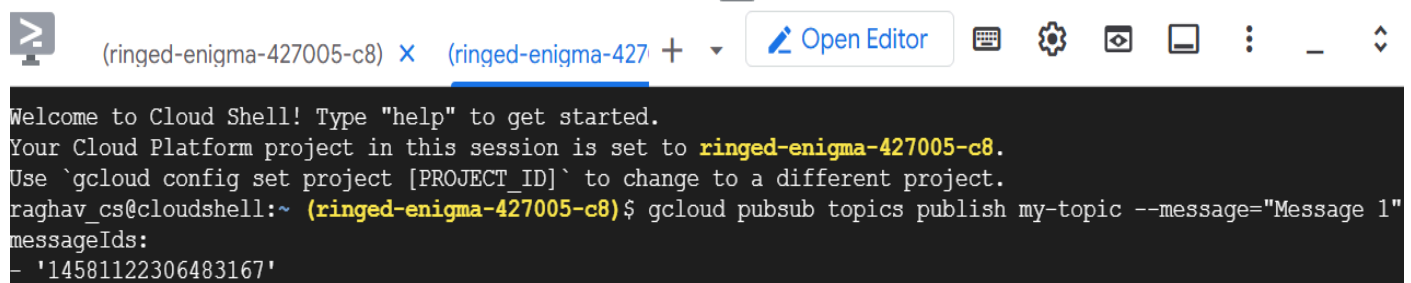
```

```

raghav_cs@cloudshell:~ (ringed-enigma-427005-c8)$ gcloud pubsub topics publish my-topic --message="Hello, Pub/S
ub!"
messageIds:
- '14581505446970720'
raghav_cs@cloudshell:~ (ringed-enigma-427005-c8)$ gcloud pubsub subscriptions pull my-subscription --auto-ack
DATA: Hello, Pub/Sub!
MESSAGE_ID: 14581505446970720
ORDERING_KEY:
ATTRIBUTES:
DELIVERY_ATTEMPT:
ACK STATUS: SUCCESS

```

Terminal 2:



```

Welcome to Cloud Shell! Type "help" to get started.
Your Cloud Platform project in this session is set to ringed-enigma-427005-c8.
Use `gcloud config set project [PROJECT_ID]` to change to a different project.
raghav_cs@cloudshell:~ (ringed-enigma-427005-c8)$ gcloud pubsub topics publish my-topic --message="Message 1"
messageIds:
- '14581122306483167'

```

Terminal 1:

```

raghav_cs@cloudshell:~ (ringed-enigma-427005-c8)$ gcloud pubsub subscriptions pull my-subscription --auto-ack -
-limit=10
DATA: Message 1
MESSAGE_ID: 14581122306483167
ORDERING_KEY:
ATTRIBUTES:
DELIVERY_ATTEMPT:
ACK STATUS: SUCCESS

```

```
raghav_cs@cloudshell:~ (ringed-enigma-427005-c8)$ gcloud pubsub subscriptions delete my-subscription
Deleted subscription [projects/ringed-enigma-427005-c8/subscriptions/my-subscription].
raghav_cs@cloudshell:~ (ringed-enigma-427005-c8)$ gcloud pubsub topics delete my-topic
Deleted topic [projects/ringed-enigma-427005-c8/topics/my-topic].
```